| Year | I Quarter | II Quarter | III Quarter | IV Quarter |
|---|---|---|---|---|
| 2005 | 255 | 351 | 425 | 400 |
| 2006 | 269 | 310 | 396 | 410 |
| 2007 | 291 | 332 | 358 | 395 |
| 2008 | 198 | 289 | 310 | 357 |
| 2009 | 200 | 290 | 331 | 359 |
| 2010 | 250 | 300 | 350 | 400 |
| Quarterly Total | 1463 | 1872 | 2170 | 2321 |
| Quarterly Averages | 243.83 | 312 | 361.67 | 386.83 |

Table 9.8

$$\text{S.I for I Quarter} = \frac{Average\ of\ I\ quarter}{Grand\ average} \times 100$$

$$\text{Grand Average} = \frac{1304.333}{4} = 326.0833$$

$$\text{S.I for I Q} = \frac{243.8333}{326.0833} \times 100 = 74.77$$

$$\text{S.I for II Q} = \frac{312}{326.0833} \times 100 = 95.68$$

$$\text{S.I for III Q} = \frac{361.6667}{326.0833} \times 100 = 110.91$$

$$\text{S.I for IV Q} = \frac{386.833}{326.0833} \times 100 = 118.63$$

## Exercise 9.1

1. Define Time series.
2. What is the need for studying time series?
3. State the uses of time series.
4. Mention the components of the time series.
5. Define secular trend.
6. Write a brief note on seasonal variations
7. Explain cyclic variations
8. Discuss about irregular variation
9. Define seasonal index.
10. Explain the method of fitting a straight line.
11. State the two normal equations used in fitting a straight line.
12. State the different methods of measuring trend.
13. Compute the average seasonal movement for the following series

| year | Quarterly Production | | | |
|---|---|---|---|---|
| | I | II | III | IV |
| 2002 | 3.5 | 3.8 | 3.7 | 3.5 |
| 2003 | 3.6 | 4.2 | 3.4 | 4.1 |
| 2004 | 3.4 | 3.9 | 3.7 | 4.2 |
| 2005 | 4.2 | 4.5 | 3.8 | 4.4 |
| 2006 | 3.9 | 4.4 | 4.2 | 4.6 |

14. The following figures relates to the profits of a commercial concern for 8 years

| Year | Profit (₹) |
|---|---|
| 1986 | 15,420 |
| 1987 | 15,470 |
| 1988 | 15,520 |
| 1989 | 21,020 |
| 1990 | 26,500 |
| 1991 | 31,950 |
| 1992 | 35,600 |
| 1993 | 34,900 |

Find the trend of profits by the method of three yearly moving averages.

15. Find the trend of production by the method of a five-yearly period of moving average for the following data:

| Year | Production ('000) |
|---|---|
| 1979 | 126 |
| 1980 | 123 |
| 1981 | 117 |
| 1982 | 128 |
| 1983 | 125 |
| 1984 | 124 |
| 1985 | 130 |



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